

KRISHNA PURWAR

AI/ML Engineer — Production AI Systems — Automation & Infrastructure

📍 Chennai, India — Open to Pan India

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Professional Summary

AI Infrastructure Engineer specializing in production RAG systems, LLM optimization, and full-stack development. Built end-to-end recruitment intelligence platform with FastAPI backend, Chrome extension ATS, and document parsing pipelines. Published 7 technical deep-dives on AI/ML topics. Proven track record of shipping production systems from 0-to-1 with expertise in vector databases, async task orchestration, and multi-LLM integration.

Education

Vellore Institute of Technology

Bachelor of Science in Computer Science & Engineering

Sep 2021 – May 2025

Chennai, Tamil Nadu

Experience

F22 Labs

Chennai, India

AI/ML Engineer

July 2025 – Present

- Engineering end-to-end **Recruitment Analytics SaaS** handling candidate profile analysis at scale with **FastAPI** backend, **React.js** frontend, **PostgreSQL** database, and **Celery** workers for async processing.
- Architected LinkedIn and Naukri scraping automation pipelines with **reverse-engineered internal APIs**, handling rate limits, anti-bot measures, and dynamic DOM parsing for candidate profile extraction.
- Built production **Chrome Extension ATS** using Manifest V3 with AI-powered candidate scoring, custom queue system with concurrency control, and IndexedDB for client-side data persistence.
- Designed **RAG-powered document analysis system** using **Qdrant** vector database, LlamaIndex hierarchical chunking, and multi-format parsing (PDF/DOCX via Gemini 2.5 Pro & LlamaParse).
- Orchestrated containerized deployments via **Docker** with CI/CD automation, ensuring 99.9% uptime for production.
- Implemented SSE streaming for real-time AI responses, reducing latency from 2-3s to sub-second token streaming.

AI/ML Intern

Feb 2025 – July 2025

- Optimized **LLMs** to achieve sub-200ms response times in real-time speech-to-speech pipeline (STT → LLM → TTS) for conversational AI product.
- Conducted systematic benchmarking of **OCR models** (Mistral OCR, GOT, OlmOCR) with quantitative accuracy metrics, published findings as technical blog with 1,000+ views.
- Developed 15+ **AI POCs** evaluating production readiness of trending Hugging Face models and GitHub repositories, reducing team's model selection time by 40%.
- Implemented **RAG pipelines** improving conversational AI accuracy by 35%.

Eternal Quill (Web App)

Aug 2024 – Feb 2025

Founder & Full-Stack Developer

Tamil Nadu, India

- Built full-stack collaborative writing platform with **React.js** frontend and **Firebase** backend, scaling to 50+ active users and 30+ independent publications.
- Reduced deployment time by 99% (manual → 2 min automated) using **GitHub Actions** CI/CD pipeline to Firebase with automated testing and rollback strategies.
- Optimized performance with lazy loading, code splitting, and intelligent caching, improving page load speed by 30% and achieving 90+ Lighthouse score.

Technical Writing & Publications

Published 7 technical deep-dives on AI/ML topics for F22 Labs engineering blog (2,500+ total views):

- **OCR Models Comparison:** Systematic benchmarking of Mistral, GOT, OlmOCR for production document processing.
- **Document Parsers Benchmark:** Comparative analysis of 5 parsing solutions (LlamaParse, Docling, PyMuPDF, etc.) with accuracy metrics.
- **Instructor vs OpenAI for Structured Output:** Guide to forcing LLMs to output consistent JSON with validation patterns.
- **Quantization Practical Guide:** Implementing 4-bit/8-bit quantization for model compression with performance tradeoffs.
- **S3 Vectors Deep Dive:** Binary quantization and vector storage optimization strategies for production RAG systems.
- **Temperature/Top_p/Top_k Explained:** LLM sampling parameters and their impact on generation quality.
- **Chrome Extension Security:** Code obfuscation techniques and reverse-engineering protection for browser extensions.

Rapid prototyping of trending AI models and tools from Hugging Face/GitHub for production evaluation.

Key Projects

Knotopian AI – Production RAG System | *FastAPI, Gemini 2.5, GPT-4.1, Qdrant, LlamaIndex* **2025**

- Built high-performance async RAG backend with **streaming responses** (SSE) for real-time document Q&A, handling multi-format uploads (PDF/DOCX/PPTX).
- Implemented **Gemini 2.5 Pro** for PDF→Markdown parsing preserving tables/hierarchy, and **LlamaParse** for Office document processing.
- Designed **Qdrant vector store** integration with OpenAI embeddings (1536 dims) and hierarchical chunking (2048/512 tokens) for optimal retrieval accuracy.
- Architected Controller-Service-Repository pattern with dependency injection for 90%+ test coverage.

LinkedIn Profile Analyzer – Chrome Extension ATS | *JavaScript, Manifest V3, IndexedDB, GPT-4o* **2025**

- Built enterprise-grade Chrome extension for automated candidate screening with LinkedIn profile scraping, resume parsing, and AI scoring.
- Engineered **Manifest V3 service worker** with custom task queue and concurrency control to handle batch processing of 50+ profiles while respecting API rate limits.
- Implemented **state machine** for analysis workflow (Pending → Parsing → Analyzing → Completed) with automatic retry logic and error recovery.
- Designed **IndexedDB schema** with Dexie.js for storing candidate profiles, PDF files (Base64), and analysis history with full-text search capabilities.
- Integrated **LlamaParse** for resume PDF extraction and **GPT-4o** with structured prompts for must-have/good-to-have/disqualifier scoring.

Reddit Summarizer – Chrome Extension | *JavaScript, Manifest V3, Groq API, XPath* **2025**

- Built intelligent Reddit comment summarizer with **multi-fallback XPath selectors** to handle Reddit's constantly changing DOM structure.
- Implemented comment ranking algorithm grouping by score and extracting top 5 score groups for quality signal extraction.
- Integrated **Groq API** (Llama 3.1) with custom prompt engineering and exponential backoff retry logic for rate limit handling.
- Designed draggable/resizable UI panel with localStorage for API key management and custom instruction support.

MobyDik – Android AI Agent POC | *Kotlin, Gemini 3 Pro, Android Accessibility* **2025**

- Developed generalist AI agent for Android using **Accessibility Services (uiautomator)** to perceive screen context and execute arbitrary tasks autonomously.
- Integrated **Gemini 3 Pro** for real-time screen understanding and action planning, solving app lifecycle challenges for background execution stability.
- Architected agentic workflow handling complex multi-step tasks (e.g., "book a cab to airport") with error recovery and adaptive replanning.

MedBot – Healthcare RAG Chatbot | *Python, Qwen 2.5 7B, RAG, Gradio* **2025**

- Built healthcare chatbot using **Qwen 2.5 7B** (7B parameters) with RAG for processing medical documents (prescriptions, lab reports, hospital records).
- Implemented custom knowledge retrieval from structured/unstructured medical data with 85% query accuracy through domain-specific fine-tuning.

Technical Skills

Languages: Python, JavaScript (ES6+), Kotlin, Java, C++, SQL, Bash

Backend & Frameworks: FastAPI (Expert), Flask, Celery, Redis, REST APIs, WebSocket/SSE, Async/Await

AI/ML & LLMs: RAG Pipelines, LLM Optimization, Multi-LLM Orchestration, Prompt Engineering, LlamaIndex, LangChain, OpenAI API, Google Gemini, Groq, Qwen, Model Fine-tuning, Quantization (4-bit/8-bit)

Vector & Databases: Qdrant (Vector Store), PostgreSQL, MySQL, Redis, IndexedDB, Firestore, Firebase Realtime DB

Cloud & DevOps: AWS (EC2, S3, Lambda, RDS - SAA Certified), Docker, GitHub Actions, Firebase, CI/CD, Linux/Unix

Front-End: React.js, Tailwind CSS, Vanilla JS, HTML5/CSS3, Chrome Extension (Manifest V3), Android XML

Tools & Platforms: Git, VS Code, Postman, Jupyter, PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, XPath

Certifications

AWS Solutions Architect - Associate (SAA-C03) | Amazon Web Services, 2024

Artificial Intelligence & Machine Learning | Powered by Google Developers

Achievements & Impact

Technical Writing | Published 7 AI/ML blog posts with 2,500+ total views, establishing thought leadership in RAG systems and document processing.

Production Systems | Shipped 12,000+ LOC recruitment intelligence platform from 0-to-1, handling candidate analysis at scale.

Community Building | Founded Eternal Quill platform with 50+ active users, 30+ publications, and 70k+ blog views in 3 months.

Investment Success | Achieved 42.60% XIRR in personal stock portfolio through data-driven quantitative strategies.

Additional Information

Interests: Technical Writing, Stock Market Trading (42.60% XIRR), Geopolitics, Poetry, Art & Cinema

Open Source: Active contributor evaluating trending AI models/repos for production readiness